Anti-theft Vehicle Monitoring and Tracking Android Application Using Firebase as Web Service

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***Abstract*—The number of vehicle users in Indonesia keep on increasing each year and this triggers probability of criminal activities especially vehicle theft. There were at least 1,688 cases of vehicle theft reported in Aceh Province throughout year 2016. Therefore, it is essentially important to keep vehicles save. One way to do it is by utilising an Android application that is capable to monitor and track the vehicles. This application has a special built-in feature: the ability to locate the position of user's vehicle with the usage of GPS and Google maps. If the vehicle moves from its original position, the application will notify the user and track its new position. Then, the vehicle owner can ask for help from friends and nearby polices. This application was developed using the Agile software methodology. Three tests were conducted for the application. The first test is the functional testing. This test was conducted to ensure that the application works in public and crowded places. The result shows that the application is functioning properly. The second test is the distance accuracy test, tested using altitude and without altitude. The results show that the error rates when altitude is used are 24.24% and 14.94% indoor and outdoor, respectively. When the altitude is not used, the error rates are 26.55% for indoor and 20.55% for outdoor. It is clear that incorporating altitude is more accurate than without using altitude. The last test is the response time between the application and Firebase as web service. We found that the average response time is 2.15 seconds which is considered good for a smartphone application.**

***Index Terms*—anti-theft application, vehicle tracking system, android, firebase, altitude**

I. Introduction

The number of vehicles keep on increasing each year. Based on data from Central Bureau of Statistics (BPS), the total of vehicles in Indonesia continues to increase from 121,394,185 in 2015 to 129,281,079 in 2016 [1]. While the number of motorcycles occupy more than 80% from all vehicles. In 2015, the number of motorcycles were 98,881,267 units and they increased to 105,150,082 in 2016 [1]. This huge number triggers the crime rate for vehicle theft to become high especially for motorcycle that are more prone to be stolen [2]. In Indonesia, BPS [3] reported that there were 38,389 vehicle theft in 2015 and 37,871 in 2016. While in Aceh province, there were 1,661 vehicle theft (or 4.32% of total vehicle theft) in 2015 and 1,688 (or 4.46% of total vehicle theft) in 2016. Vehicle theft causes loses both material and psychological to the victim [4].

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This increasing vehicle theft causes people restless and reluctant to leave the vehicle outside of their supervision. These recent years with the help of technology, there are some surveillance ways to prevent vehicle theft. For example using remote alarm system, Closed-Circuit Television (CCTV), smartphone and anti-theft device or system. Just like its name, smartphone is smart with the help of many applications that can be installed on it. Smartphone operates by using operating system such as Android, iOS, Windows. The major smartphone market uses Android as its operating system because Android is open source so anyone can develop any kind of application for Android smartphone for free. Android smartphone has Global Positioning System (GPS) installed so Android smartphone can be used to track location. So this research focuses on developing prototype for anti-theft vehicle Android application.

II. Related Works

Researchers have been researched anti-theft device or system that can help prevent, monitor and track vehicle theft. Many of those researches focuses on the hardware part of the device or system. For example [5] uses Radio Frequency Identification (RFID) technology to link the driver's key and the ignition system so if the key with ignition system is not present, the engine will not start despite the keylock has been bypassed or broken. RFID technology was also used by [6]. Zhixiong [7] uses computer vision technology to locate and recognize the driver's face, identify the unauthorized driver. It uses active IR illuminator to acquire driver's images in real time so that when unauthorized driver is driving, it will alarm and send the unauthorized driver's image to the car owner or police through CDMA (Code Division Multiple Access) or GPRS (General Packet Radio Service) networks. Another protection method is by using sensor to verify driver by analyzing driving pattern [8]. This mechanical feature can differentiate drivers' driving behaviors. Fingerprint sensor can also be used to verify the driver's identity [9]. The fingerprint should be taken first before starting engine. If vehicle's location is changed without fingerprint verification, a message will be sent to owner's handphone.

Anti-theft device or system mostly uses the combination of Global Position System (GPS) and Global System for Mobile

(GSM). GPS is used to locate the vehicle location while GSM is used to communicate with the vehicle for easier finding after a theft attempt [10]. The device or system is mostly made by combining a microcontroller with GPS and GSM [10], [11], [12], [9], [13], [6]. Other than combining GPS, GSM and microcontroller, [6] and [14] added RFID technology to the anti-theft device or system.

This recent year, researchers also utilize smartphone in their research of anti-theft device or system. The smartphone being used is mostly Android-based smartphone. Liu [14] developed anti-theft system that combine RFID, GPS and GSM module to lock, unlock and cut off gasoline. The system will send the location information obtained by GPS to owner's mobile phone when the car is stolen. The information is sent in SMS (Short Message Service) format that includes longitude and latitude of stolen car. Monawar [15] proposed a Vehicle Tracking System (VTS) that is pretty much the same, but it has database to store police station's location and number. The system will calculate the nearest police station by using Haversine formula.

Lee [16] developed VTS by using microcontroller that has GPS and GSM/GPRS module. GPS module gets the geographic coordinate and GSM/GPRS module transmit and update the vehicle location to a database. After that an iOS smartphone application continously monitors the vehicle and displays the vehicle on Google Maps. Shruti [17] designed and developed pretty similar VTS with [16]. But [17] uses 2 microcontrollers and 1 Raspberry Pi. One of microcontroller has GPS and GSM modules to get geographic coordinate and send the information to user's smartphone application. The other microcontroller and Raspberry Pi are used to capture image and send email with the attachment of the thief's capture face to user's email account. User will also get an SMS alert and later can direct the SMS alert and email to the police control room. [17] developed Android application to save vehicle information, track and show location on map, get and send email to police.

A smart anti-theft VTS was developed by [18]. Just like the other VTS, it also has tracking/monitoring function. But [18] has added location saving function to Amazon web service so the VTS can be an Internet of Things. The device and system developed by [18] also has fuel control and driver authorization to do emergency stop by closing the fuel line. This function can be controlled from a smartphone application. Most anti-theft VTS researches discussed previously are intended for car stealing, Hossain [19] proposed a design of low cost anti-theft sensor for motorcycle. The sensor will detect the movement of handlebar and will send a warning SMS to the owner.

III. Proposed Methodology

Many anti-theft vehicle application are developed by combining microcontroller that has GPS and GSM capability and other monitoring device such as website, SMS and mobile application. This research proposed an anti-theft vehicle application that can also act as VTS. The application is based on Android smartphone because Android-based smartphone is the most widely mobile Operating System usage in the world [20]. The proposed application will only require Android smartphone compared with other previous researches that require more tools.

The proposed anti-theft VTS consists of 2 parts, main application and vehicle application. The main application will have 2 kinds of login. The first one is normal user and the second one is the police user. The normal user can track and monitor his/her vehicle on Google Maps. So if there is any movement made to the vehicle within some distances, owner will get notification. Once owner get notification, owner can ask for help by sending email and notification to friends and nearest polices. The vehicle application will update the vehicle's location on cloud database (CDB) in real time. So if there's any movement in certain distance, the application in vehicle will immediately send notification to main application in owner's smartphone. The architecture of the proposed anti-theft VTS can be seen at Fig 1.

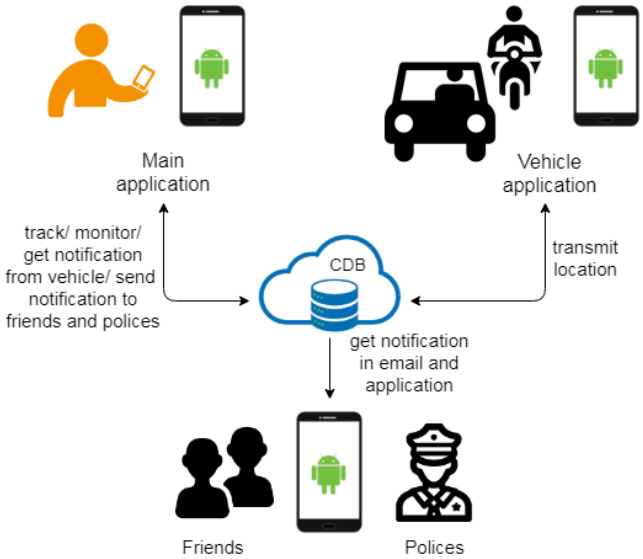

Fig. 1. Proposed anti-theft VTS architecture.

This research mainly consists of application development and testing. The flow process, as seen in Fig 2, starts with the development of vehicle application with Android Studio. The application will have function to find vehicle's current location consists of latitude, longitude and altitude data. Then the application will have function to immediately transmit location data to cloud database. While in the other part, main application in user's smartphone will be also developed using Android Studio. The main application will have login function for normal user and police user. The main application will have function to automatically get the vehicle location data from cloud database. The data will be used to calculate distance using latitude and longitude information. The distance will also be tested with altitude information. The application will also have notification and sharing function. Once both application prototype are ready, they will be tested.

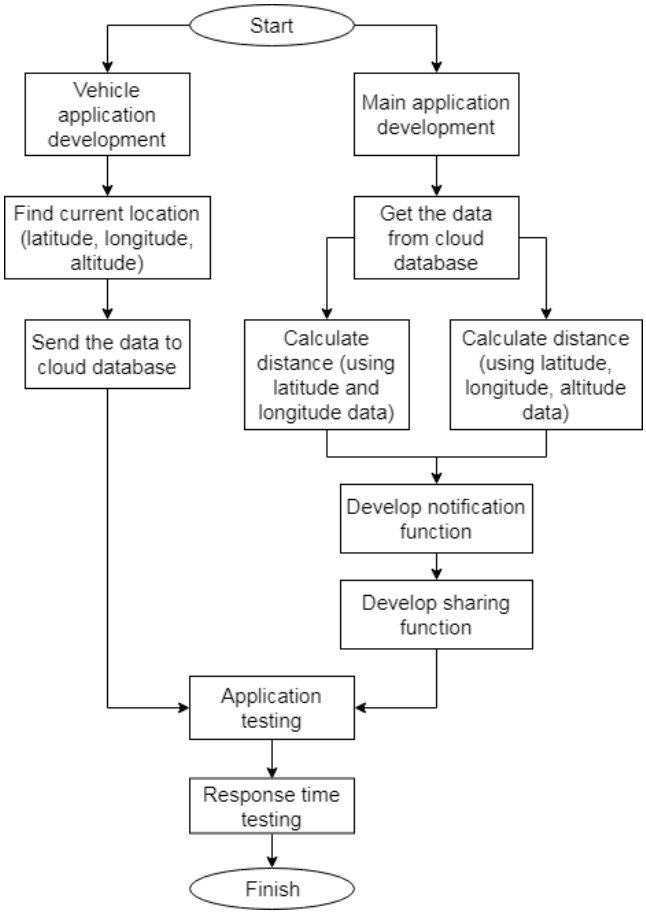

Fig. 2. Research flow process.

The application will be developed by following Agile software engineering methodology. Agile methodology is targeted to small and medium-sized software projects. Agile methodology focuses on working software so it emphasizes development team to write code not write document. Agile methodology enable developer to adjust the changing requirement quickly and minimize any mistake during development process [21].

IV. Design and Implementation

The application is designed for Android-based smartphone and developed with Android Studio software. Android Studio has Gradle-based build system and built-in support for Google Cloud Platform. One of Google Cloud Platform is Firebase. Firebase is a cloud storage for any kind of application in various devices. Firebase provides realtime database that can store and synchronize application data in seconds as long as there is internet connection. Firebase is a NoSQL database that stores data as JSON format. This application's implementation uses Firebase to store vehicle's location and user. It also stores other additional data such as email, battery status, friends list and polices list. The use of Firebase as cloud database enable both vehicle and user applications to access data in real time.

Anti-theft VTS in this research has 2 parts, they are vehicle application and main application. Vehicle application is the application that will be put in vehicle, either car or motorcycle. This application collects and update data about the current vehicle location then transmit the data to Firebase in real time as long as the application is connected to internet and has enough battery power. This application will also update battery status data in the device and transmit to Firebase.

The main application will be used by user or owner. This application is designed with these features:

- Tracing the vehicle and monitor it on Google map.
- The distance between vehicle and owner can be set by the owner as a threshold distance so when the distance change by itself, the application will notify the owner.
- Owner can ask for help from his/her friends who were already added to friends list. Other than friend, owner can also send email to nearest police officer.
- User can monitor user's friend vehicle so he/she can track and find the vehicle if his/her friend's vehicle was stolen.

Both application have some common functions like registration, login, update location and battery status. Whereas main application has additional function like calculating distance, displaying map, getting notification. The functions will be explained in more detail below.

A. Registration

Registration is a function for user to register active email and password. Other compulsory data recorded is name, vehicle's brand, vehicle's license plate. All data will be stored in Firebase. The purpose of this registration is to differentiate each user in the application.

User who register as police officer will need additional data which is a permit from regional police chief. This permit should be in pdf file uploaded during registration. The file will be checked whether it has more than 25 characters in it. This checking to minimize unauthorized person registering as police officer. But this solution is only temporary. For future research, the permit will be fully checked by using Optical Character Recognition.

B. Login

All user need to login on the application by inputting email and password used during registration. This part will be synchronized with the data in Firebase. This login function is required for tracking and monitoring user's vehicle.

C. Update location and battery status

Both application will update current location and battery status in real time to Firebase. The location data consists of latitude, longitude and altitude. The data is taken from method $getLatitude()$, $getLongitude()$ and $getAltitude()$ using installed GPS in smartphone. To ensure the location data is accurate, the data was manually compared with the real calibrated GPS data. The result from this calibration shows that there is 24 meter difference in altitude data. So altitude data needs to be added by 24 to give correct result in equation used in application.

D. Calculating distance

Calculating distance function needs latitude, longitude and altitude. There are 2 types of calculating distance, they are by using altitude and without using altitude. To calculate distance with altitude, latitude and longitude coordinates are converted to meter. 1 degree latitude is equal to 110,574 meter. 1 degree longitude is equal to $111,320 * cos(latitude) * 1000$. After that the distance is calculated by using three-dimensional Cartesian in equation 1. Calculating distance without altitude is easier since it can use method $distanceTo()$. The display for information about distance can be seen in Fig 3.

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2} \quad (1)$$

Fig. 3. Distance information interface.

E. Displaying map

Map is displayed with the help of Google Maps Activity that will show vehicle and user in Google Maps. Marker is used to pinpoint the vehicle and user in Google Maps. Method addCircle() is used to display red circle on map based on the distance alredy set by user. The map showing vehicle and owner can be seen in Fig 4.

F. Getting notification

Notification will pop up in owner's smartphone when the distance between vehicle and owner is more than the distance set by owner. Notification means the vehicle is moving away from owner or the vehicle is in danger. If owner gets notification, owner can send notify and email his/her friends on list or nearest police officers. Then owner's friend or police can view the vehicle on Google map. A notification example can be seen in Fig 5.

Fig. 4. Map interface.

Fig. 5. Notification shown when vehicle is out of circle.

V. Result and Discussion

There were three tests conducted after the development of this application. The first test conducted was functional testing. This test was conducted to ensure that the application works in public places such as in Syiah Kuala University and crowded places in Lampineung area. The test in Syiah Kuala university uses 2 Android smartphone, one is installed with vehicle application and the other has main/user application installed. This test was done successfully. The test around Lampineung area involves 4 Android smartphones. One is in vehicle, second is owner, third is owner's friend and fourth is police officer. This test was also done successfully. At the end of this test, it is proven that this application is functioning properly.

The second test conducted was the distance accuracy test, both using altitude and without using altitude. The test was done indoor and outdoor. First the real distance was measured by using metric tape measure. Then the distance shown in application was noted down. To ensure the distance accuracy, percent error formula was calculated for each result. Percent

error is a useful formula for determining the precision of calculation. The formula for percent error is in equation 2. The value of measurement is considered more accurate if the percent error value close to zero [22].

$$\%error = |\frac{calculateddistance - exactdistance}{exactdistance}| * 100 \quad (2)$$

For indoor test, real distance is measure with tape measure is 7.9 meters (using altitude) and 7.45 meters (without altitude). The result of distance calculation and percent error are shown in Table I. As for outdoor test, real distance measurement is 10.6 meters (using altitude) and 10 meters (without altitude). The result of distance calculation and percent error are shown in Tabel II.

TABLE I
DISTANCE COMPARISON AND PERCENT ERROR FOR INDOOR TEST

Distance Calculated by Application (meter)		Percent Error (%)	
Using Altitude	Without Altitude	Using Altitude	Without Altitude
9.6	9.27	21.52	24.43
8.07	7.12	2.15	4.43
9.44	8.11	19.49	8.86
6.06	5.69	23.29	23.62
9.45	13.33	19.62	78.93
12.41	13.18	57.09	76.91
11.63	9.18	47.22	23.22
6.74	7.01	14.68	5.91
6.53	6.66	17.34	10.6
6.32	6.81	20	8.58
Average		24.24	26.55

TABLE II
DISTANCE COMPARISON AND PERCENT ERROR FOR OUTDOOR TEST

Distance Calculated by Application (meter)		Percent Error (%)	
Using Altitude	Without Altitude	Using Altitude	Without Altitude
12.02	12.59	13.4	25.9
11.58	12.22	9.25	22.2
9.94	10.46	6.23	4.6
9.27	9.12	12.55	8.8
9.1	8.3	14.15	17
7.94	8.11	25.09	18.9
8.12	8.07	23.4	19.3
8.73	7.78	17.64	22.2
12.75	6.56	20.28	34.4
10.04	9.6	7.36	32.2
Average		14.94	20.55

From the test, data error percentages were collected. The results using altitude were 24.24% error indoor and 14.94% error outdoor rates. While the results without using altitude were 26.55% error indoor and 20.55% error outdoor rates. The data showed that using altitude to determine the distance was more accurate compared to without using altitude.

The last test was conducted to application to know response time between application and Firebase as the web service. This test used 3 accounts for both vehicle and main applications. The total of smartphone being used is 6 smartphones, 3 smartphones used vehicle application and the other 3 smartphones used main application. Device brand and telecommunication provider were ignored in this test. So this test focused on response time between application and Firebase and ignored the impact of device brand and telecommunication provider. The result detail for this test is shown in Table III. The result of this test was that the average response time of the application was 2.15 seconds. This response time is good for a smartphone application. According to [23], application on smartphone is considered poor responsiveness if the response time exceeds 5 seconds because an "Application Not Responding" (ANR) error will be displayed to the user. Such error creates negative perception to user.

TABLE III
THE RESULT OF RESPONSE TIME TESTING

Testing	1st response time (second)	2nd response time (second)
Test no. 1	0	1
Test no. 2	1	5
Test no. 3	2	5
Test no. 4	2	4
Test no. 5	1	3
Test no. 6	0	3
Test no. 7	0	3
Test no. 8	1	3
Test no. 9	0	4
Test no. 10	1	4
Response time average	2.15	

VI. CONCLUSION AND FUTURE WORK

Anti-theft VTS for Android smartphone was developed by following Agile software methodology. The application consists of 2 parts, vehicle application and main application. Both application store and update data in Firebase as cloud database by using web service. Vehicle application is the application that will be put in vehicle, either car or motorcycle. This application collects and update data about the current vehicle location then transmit the data to Firebase in real time. Main application is for user or owner to track and monitor the vehicle. The application can show notification if the vehicle move away and out of specified distance. So user can ask for help from friends and police officers. There are three type of testing being conducted. The first one is functional testing. This test was conducted to ensure that the application works in public places such as in Syiah Kuala University and crowded places in Lampineung area. The second test conducted was the distance accuracy test, both using altitude and without using altitude. From the test, data error percentages were collected. The results using altitude were 24.24% error indoor and 14.94% error outdoor rates. While the results without using altitude were 26.55% error indoor and 20.55% error outdoor rates. The percent error data showed that using altitude to measure the distance was more accurate compared to without using altitude. The last test conducted was from Firebase as the web service. The result of this test was that the average response time of the application was 2.15 seconds. This response time has good responsiveness for a smartphone application.

This research focused on anti-theft VTS for Android smartphone. So in future research. the application should be targeted to other mobile operating system. Also the vehicle application should be replaced with a microcontroller for example Arduino or Raspberry Pi that is installed with GPS tracker. This way the tracker can be permanently mounted in the vehicle. The application should also add other feature for example uploading profile photo for verification and security reason. Another feature is rating for police officer who successfully caught the vehicle thief. Also the important thing is adding a more secure verification for police officer to ensure the police officer is a real police. The future improvement should also measure the speed of vehicle so that it can still be traced by the application also measure the signal strength of the smartphone.

REFERENCES

[1] B. P. Statistik, "Perkembangan Jumlah Kendaraan Bermotor Menurut Jenis , 1949-2016."

[2] A. M. Alfiyansyah, "Tinjauan Kriminologis Terhadap Kejahatan Pencurian Kendaraan Bermotor," Ph.D. dissertation, Hasanuddin University, 2013.

[3] B. P. Statistik, "Statistik Kriminal 2017," Badan Pusat Statistik, Tech. Rep., 2017.

[4] C. Q. Akyunin, "Kajian Viktimologi Tentang Perlindungan Hukum Bagi Korban Pengendara Kendaraan Bermotor yang Mengalami Tindak Pidana Pecurian dengan Kekerasan," *Sarjana Ilmu Hukum*, no. April, 2013.

[5] K. Khangura, N. Middleton, and M. Ollivier, "Vehicle anti-theft system uses radio frequency identification," in *IEE Colloquium on Vehicle Security Systems*, 1993, pp. 4–7.

[6] M. Geetha, T. Priyadarshini, B. Sangeetha, and S. Sanjana, "Anti-theft and tracking mechanism for vehicles using GSM and GPS," in *2017 Third International Conference on Science Technology Engineering & Management (ICONSTEM)*, 2018, pp. 252–255.

[7] L. Zhixiong and H. Guiming, "A vehicle anti-theft and alarm system based on computer vision," in *2005 IEEE International Conference on Vehicular Electronics and Safety Proceedings*, 2005, pp. 326–330.

[8] B. I. Kwak, J. Y. Woo, and H. K. Kim, "Know your master: Driver profiling-based anti-theft method," in *2016 14th Annual Conference on Privacy, Security and Trust, PST 2016*, 2016, pp. 211–218.

[9] M. Dey, M. A. Arif, and M. A. Mahmud, "Anti-Theft protection of vehicle by GSM & GPS with fingerprint verification," in *ECCE 2017 - International Conference on Electrical, Computer and Communication Engineering*, 2017, pp. 916–920.

[10] M. F. Saaid, M. A. Kamaludin, and M. S. Ali, "Vehicle location finder using Global position system and global system for mobile," in *Proceedings - 2014 5th IEEE Control and System Graduate Research Colloquium, ICSGRC 2014*, 2014, pp. 279–284.

[11] K. Mukherjee, "Anti-theft vehicle tracking and immobilization system," in *2014 International Conference on Power, Control and Embedded Systems*, 2014, pp. 1–4.

[12] K. A. Mamun and Z. Ashraf, "Anti-theft vehicle security system with preventive action," in *2015 2nd Asia-Pacific World Congress on Computer Science and Engineering*, 2015, pp. 1–6.

[13] A. Wu, "The design of anti-theft device for vehicle based on GSM," in *2017 International Conference on Smart Grid and Electrical Automation*, 2017, pp. 384–386.

[14] Z. Liu, A. Zhang, and S. Li, "Vehicle anti-theft tracking system based on Internet of things," in *Proceedings of 2013 IEEE International Conference on Vehicular Electronics and Safety*, 2013, pp. 48–52. [Online]. Available: http://ieeexplore.ieee.org/document/6619601/

[15] T. Monawar, S. B. Mahmud, and A. Hira, "Anti-theft vehicle tracking and regaining system with automatic police notifying using Haversine formula," in *2017 4th International Conference on Advances in Electrical Engineering (ICAEE)*, 2017, pp. 775–779.

[16] S. Lee, G. Tewolde, and J. Kwon, "Design and implementation of vehicle tracking system using GPS/GSM/GPRS technology and smartphone application," in *IEEE World Forum on Internet of Things, WF-IoT 2014*, 2014, pp. 353–358.

[17] K. Shruthi, P. Ramaprasad, R. Ray, M. A. Naik, and S. Pansari, "Design of an anti-theft vehicle tracking system with a smartphone application," in *IEEE International Conference on Information Processing, ICIP 2015*, 2015, pp. 755–760.

[18] M. S. Uddin, M. M. Ahmed, J. B. Alam, and M. Islam, "Smart anti-theft vehicle tracking system for Bangladesh based on Internet of Things," in *4th International Conference on Advances in Electrical Engineering, ICAEE 2017*, 2017, pp. 624–628.

[19] M. M. Hossain, M. S. Islam, N. F. Dipu, M. T. Islam, S. A. Fattah, and C. Shahnaz, "Design of a low cost anti-theft sensor for motorcycle security device," in *5th IEEE Region 10 Humanitarian Technology Conference 2017, R10-HTC 2017*, 2017, pp. 778–783.

[20] Statista Inc., "Global smartphone sales to end users from 1st quarter 2009 to 1st quarter 2018, by operating system (in million units)," 2018. [Online]. Available: http://www.statista.com/statistics/266219/global-smartphone-sales-since-1st-quarter-2009-by-operating-system/

[21] J. Dooley, *Software development and professional practice*. New York: Apress, 2017.

[22] A. M. Helmenstine, "How to Calculate Percent Error," 2018. [Online]. Available: https://www.thoughtco.com/how-to-calculate-percent-error-609584

[23] S. Yang, D. Yan, and A. Rountev, "Testing for poor responsiveness in Android applications," in *1st International Workshop on the Engineering of Mobile-Enabled Systems (MOBS)*, 2013, pp. 1–6.